From Social Media to Wall Street: Trump Posts vs S&P 500

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Github repository

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Abstract

This paper investigates whether Donald Trump's social media posts contain signals predictive of S&P 500 movements. Drawing on a corpus of 90,551 posts spanning 2009 to 2026 across X (formerly Twitter) and Truth Social, we apply a multilayered NLP pipeline combining VADER sentiment analysis, FinBERT financial sentiment scoring, BERTopic-based topic modelling, and sentence embeddings with semantic clustering. Features are aggregated daily and merged with S&P 500 return data for modelling. We find that Trump's posts contain a modest directional signal for next-day market movements: the best classification model reaches 58.0% test accuracy, 54.0% balanced accuracy, and a ROC-AUC of 0.586. A large-move walk-forward setup reaches a ROC-AUC of 0.557, while event-day filtering adds no robust improvement. Sentiment dispersion within a day and text-structure features outperform mean sentiment as predictors, while return magnitude is not predictable under any specification. These results contribute to a growing literature on the financial relevance of political communication on social media.

Keywords: Sentiment Analysis, NLP, Social Media, S&P 500, Stock Market Prediction, VADER, FinBERT, Sentence Embeddings, Topic Modelling, Donald Trump, X, Twitter, Truth Social

Introduction

Social media has fundamentally changed how political leaders communicate with both the public and markets. Where information once flowed through press releases and news reporting subject to editorial delay, posts on X or Truth Social reach millions of followers instantaneously. Donald Trump represents a particularly salient case. His prolific posting history, spanning over a decade and two presidencies, has been linked to measurable movements in equity markets, Treasury yields, and individual stock prices. JP Morgan's Volfefe Index (named after Trump's enigmatic "covfefe" post) provided early practitioner evidence that specific words in Trump's posts were associated with Treasury yield volatility (Cembalest & Iyer, 2019), and a growing body of academic work has since confirmed a statistically meaningful, if modest, relationship between his posting behaviour and market outcomes.

This paper asks a focused question: do features derived from Trump's social media post, including sentiment, topic, posting intensity, and semantic novelty, contain information predictive of S&P 500 next-day returns? We address this through a pipeline that moves from raw text to a model-ready feature matrix, covering preprocessing, feature engineering, exploratory data analysis, and formal modelling. Our dataset is distinctive in spanning both X and Truth Social across 2009 to 2026, capturing Trump's full posting history across both platforms and both presidential terms.

We frame our prediction task within the noise trader and policy uncertainty literature. If Trump's posts shift investor sentiment or signal policy direction, they should be partially reflected in next-day index returns before markets fully adjust; a testable implication of the semi-strong form of the efficient market hypothesis. We treat classification performance above random baseline as evidence of limited directional predictability, while our regression null result is consistent with return magnitude being difficult to predict.

The remainder of the paper is organised as follows: Section 2 reviews related work. Section 3 describes the data and preprocessing. Section 4 presents the feature engineering methodology. Section 5 reports exploratory findings. Section 6 presents modelling strategy and results. Section 7 discusses implications and limitations. Section 8 concludes.

Related Work

The closest predecessor to our work is Frydendahl and Stenger (2019), a CBS Master's thesis using an event study to examine how Trump's company-specific tweets affect the stock prices of targeted firms. They find that positive tweets generate positive cumulative average abnormal returns on the tweet day, while negative tweets have a delayed negative effect. However, their study only covers the period up to May 2019 (less than half of Trump's first presidency) and focuses on firm-level stock reactions rather than the broader S&P 500. It also relies on manually coded tweet polarity instead of automated NLP-based sentiment or topic analysis. Our study addresses these gaps by covering 2009 to 2026 and applying a full NLP pipeline to extract sentiment, topic, and semantic features automatically.

Li et al. (2023) take a different angle, using Trump's Twitter archive to study misinformation dynamics during the 2020 election via synthetic control methods. They demonstrate that Trump's posts directly steer the volume and direction of election-related discourse, with tweet volume rising sharply in the 120 minutes following a Trump post. While they confirm the real-time influence of Trump's communication, their study is limited to the 2020 election period and does not apply sentiment analysis or NLP to the post content itself.

On the broader market side, Molnár et al. (2021) apply topic modelling to Trump's tweets and find that trade-war-related posts have the strongest negative effect on U.S. equities, directly motivating our focus on trade and economic topics in the `BERTopic`-derived feature set. Peric Ortiz (2023) uses an event study to show that tweets about foreign policy, monetary policy, and immigration significantly increase market uncertainty, and further finds that posting frequency itself matters independently of content.

For methodological framing, we draw on Kim et al. (2021), who examine the relationship between Elon Musk's Twitter activity and Tesla's stock price using an EDA-first approach, characterising posting patterns, sentiment distributions, and engagement statistics before formal modelling. This structure we replicate here.

Data and Preprocessing

Data Sources

Our dataset combines two sources. The primary corpus consists of 90,551 posts by Donald Trump drawn from the Trump Twitter Archive (Zlotowski, 2026), which contains his complete Twitter/X posting history, alongside his Truth Social posts scraped from his official *@realDonaldTrump* account. The combined dataset covers the period from May 2009, when Trump's account was created, through January 2026. Market data is drawn from *Yahoo Finance*, providing daily S&P 500 open, high, low, close, volume, and pre-computed daily returns for the same period.

The tweet dataset contains 18 columns including post identifier, timestamp, platform, text, favourite count, repost count, boolean flags for quote posts, reposts, and deletions, word count, extracted hashtags, URLs, user mentions, media counts, and reply-to fields. The S&P 500 dataset contains 10 columns: date, close, high, low, open, volume, daily return, absolute return, intraday range percentage, and next-day return (pre-shifted for the modelling step).

Preprocessing

Raw tweet text was preprocessed in several stages. First, all timestamps were converted to US Eastern timezone and normalised to tz-naive datetime objects to facilitate consistent merging with market data. Second, reposts were excluded from all feature computation, retaining only original posts attributed directly to Trump (74,742 of the 90,551 total). This decision follows Kim et al. (2021) and is motivated by the goal of capturing Trump's own words rather than amplified third-party content.

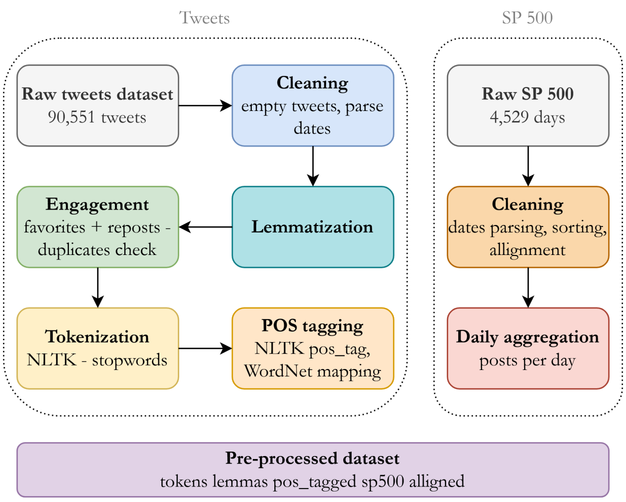

Figure 1: Overview of the feature preprocessing pipeline.

Text preprocessing for downstream NLP tasks included tokenisation using NLTK's word tokeniser, part-of-speech tagging, and POS-aware lemmatisation using WordNetLemmatizer, as seen in *Figure 1*. Stopwords from NLTK's English stopword list were removed for frequency analysis and word cloud generation. URLs, special characters, and retweet markers ("RT") were stripped where relevant. For sentiment analysis and embedding computation, raw text was retained to preserve punctuation, capitalisation, and emphasis patterns, all of which carry sentiment-relevant information for `VADER`.

The S&P 500 data required minimal preprocessing. Column names were standardised from lowercase to mixed case, and the pre-computed next-day return column (`next_day_return`) was used directly as the regression target to avoid look-ahead bias: tweet features observed on day `t` are always matched to market outcomes on day `t+1`.

We note that the direction of causation cannot be established from correlational modelling alone: Trump may post more negatively in response to market declines. Our design controls for this partially by predicting *next-day* returns from *today's* posts, but a rigorous causal identification strategy remains a direction for future work.

Methodology

Feature Engineering Pipeline

Feature engineering proceeded in three stages: tweet-level feature computation, daily aggregation, and final merge with market data. The architecture is summarised in *Figure 2*.

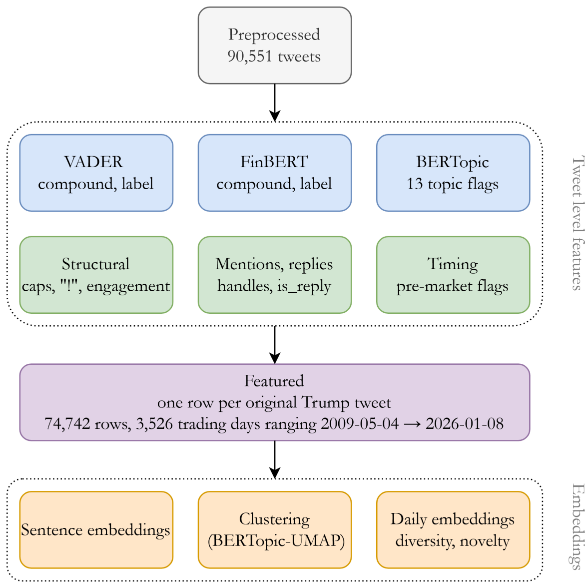

Figure 2: Overview of the feature engineering pipeline from raw posts to model-ready daily feature matrix.

Sentiment Analysis

We apply two complementary sentiment models to each original post. `VADER` (Valence Aware Dictionary and sEntiment Reasoner; Hutto & Gilbert, 2014) is a rule-based model that assigns four scores per post: positive proportion, negative proportion, neutral proportion, and a compound score in [-1, +1]. `VADER` is particularly well-suited to social media text because it incorporates rules for capitalisation, punctuation emphasis, and common social media tokens. Each post receives a compound score, and a categorical label (positive if compound $\geq$ 0.05, negative if $\leq$ -0.05, neutral otherwise) using standard cutoffs.

`FinBERT` (Araci, 2019) is a BERT-based transformer fine-tuned on financial news text. We use the ProsusAI/finbert checkpoint from HuggingFace, which classifies text as positive, negative, or neutral and outputs probabilities for all three classes. We compute a polarity score as positive probability minus negative probability, giving a continuous signal in [-1, +1].

The two models capture different aspects of the text. `VADER` is sensitive to rhetorical style such as capitalisation, exclamation marks, and superlatives, while `FinBERT` captures domain-specific financial connotations.

Topic Detection

Topic detection was performed using `BERTopic` (Grootendorst, 2022), a transformer-based topic modelling framework that combines sentence embeddings with `HDBSCAN` clustering and `c-TF-IDF` term scoring to identify coherent topics from text. Rather than defining topics by hand through keyword lists, `BERTopic` discovers them automatically from the corpus, making the approach data-driven and replicable.

Each of the 74,742 original posts was assigned to a topic by the fitted `BERTopic` model. Posts not assigned to any coherent cluster (labelled topic −1 by `HDBSCAN`) were treated as outliers and excluded from topic-based features. The remaining topics were inspected visually using bar charts of top `c-TF-IDF` terms per topic, and human-readable labels were assigned manually after group inspection. Eleven topics were retained and mapped to binary flag columns; three additional topics were left unmapped as noise (for example, tweets composed uniquely by “`[Image]`”) or too *niche* for meaningful interpretation *(Figure A1)*. Topics which are referenced the most in his posts include: Trump’s identity (13.848), Elections (7778) and Rallies/Nationalism (5786). More in depth analysis in *Table A1.*

For modelling, topic flags are encoded as binary variables and aggregated to the daily level. A day is marked as topic-relevant if at least one post belongs to that topic. The two market-relevant topics, `topic_trade_geopolitics` and `topic_economy_markets`, define “event days”, which are used in *Section 6* to test whether predictive signals are stronger when Trump posts about economically relevant content.

Structural and Stylistic Features

Beyond sentiment and topic, we compute structural features that capture Trump’s rhetorical style and posting behaviour. These include: caps ratio (fraction of alphabetic characters that are uppercase, measuring shouting intensity), exclamation count, question mark count, URL presence flag, log-transformed engagement score (log1p of favourites plus reposts), and binary timing flags indicating whether a post was sent pre-market (04:00–09:29 ET), during market hours (09:30–15:59 ET), or after hours. We also extract all *@mentioned* handles into a list column and flag reply posts using the `in_reply_to` field.

Sentence Embeddings and Novelty

To capture semantic drift in Trump's posting behaviour over time, we compute a daily mean embedding by averaging the sentence embeddings of all original posts on a given day into a single 384-dimensional vector. The novelty score for each trading day is then defined as the cosine distance between that day's mean embedding and the mean of the preceding 14 days' daily mean embeddings. The 14-day window was selected by comparing windows of 7, 14, and 30 days; while the 7-day window produces a marginally more autocorrelated signal (0.494 vs 0.466), the 14-day baseline was preferred as it is less sensitive to short-term posting bursts and provides a more interpretable two-week rhetorical baseline. A high novelty score indicates that Trump's posts on that day are semantically distant from his recent baseline, that is, he is posting about topics or using language that departs meaningfully from his typical pattern over the previous two weeks.

Daily Aggregation and Feature Matrix Construction

All tweet-level features are aggregated to the daily level. For continuous features (VADER compound, FinBERT polarity, caps ratio, exclamation count, log engagement), we compute the daily mean, standard deviation, maximum, and minimum. For binary topic flags, we take the daily maximum (1 if any post that day mentioned the topic, 0 otherwise). Post count and mention count are summed. We additionally construct a pre-market subset and compute all aggregations separately for posts sent before 09:30 ET, appending these as separate columns with a _premarket suffix.

Rolling window features are then computed at the daily level: 3-day and 7-day rolling means of `VADER` compound, and a sentiment momentum signal defined as today's `VADER` mean minus the 3-day rolling mean. The daily mean embedding is also computed per day as the mean across all original posts, and the novelty score is derived from this daily mean as described above.

The final feature matrix is constructed by merging the daily tweet feature table with S&P 500 market data on the date key. All tweet features on day `t` are matched to market outcomes on day `t+1` via the pre-shifted `next_log_return` and `next_market_up` columns, enforcing the no-look-ahead constraint. The resulting feature matrix contains approximately 3,526 trading-day rows and covers all days where both tweet data and market data are available.

Exploratory Data Analysis

Post Volume Over Time

Trump's posting activity shows a clear upward trend over the dataset period, with notable disruptions corresponding to key platform events. *Figure 3* shows annual post counts. Twitter activity peaked during his first presidency (2017–2021), with posting volume roughly doubling between 2016 and 2019. The January 2021 Twitter ban is clearly visible as a structural break, after which volume drops to near zero on that platform. Truth Social activity increases from 2022 onward, partially compensating for the Twitter loss, though the two platforms exhibit different posting cadences and audience characteristics.

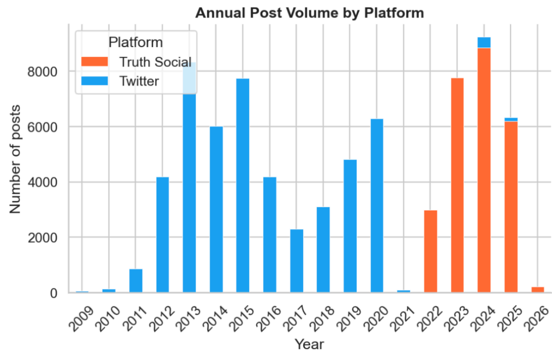

Figure 3: Annual number of original posts by Donald Trump across X (Twitter) and Truth Social.

Figure 4 shows the monthly post volume over the full timeline, revealing several notable peaks: a concentration of high-volume months in 2019–2020 (corresponding to the impeachment proceedings and COVID-19 pandemic), a sharp drop in early 2021, and a gradual recovery through Truth Social from 2022. The platform composition shifts are visible as the blue line (Twitter) gives way to red (Truth Social) after 2021.

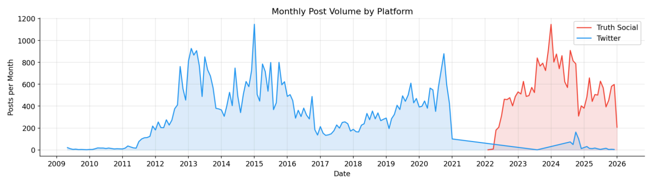

Figure 4: Monthly original post volume split by platform (X = blue, Truth Social = red).

Analysis of posting hour distribution *(Figure 5)* reveals that Trump posts most heavily in the afternoon and evening (14:00–23:00 ET), with a secondary peak in the early morning. Approximately 12% of posts fall in the pre-market window (04:00–09:29 ET), a segment of particular interest for market impact analysis, as these posts are visible to investors before the NYSE opens.

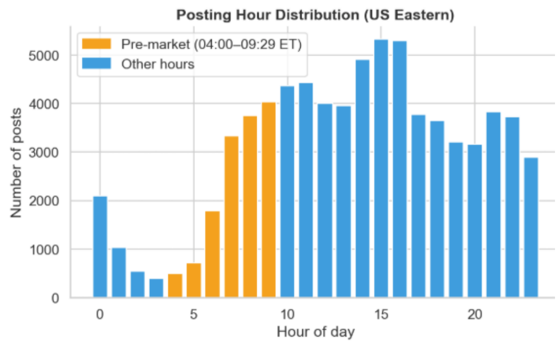

Figure 5: Distribution of posts by hour of day.

Engagement Statistics

Trump's engagement metrics show a clear structural shift around 2018, similar to patterns noted by Kim et al. for Elon Musk. From 2009 to 2017, monthly likes and reposts remained relatively stable, but increased sharply from 2018 onward, likely reflecting both his rising political profile and the expansion of Twitter (*Figure 6*). Engagement is also highly skewed, with a small share of posts generating most interactions. To address this, we use a log transformation (log1p of likes and reposts) to normalise the distribution for modeling.

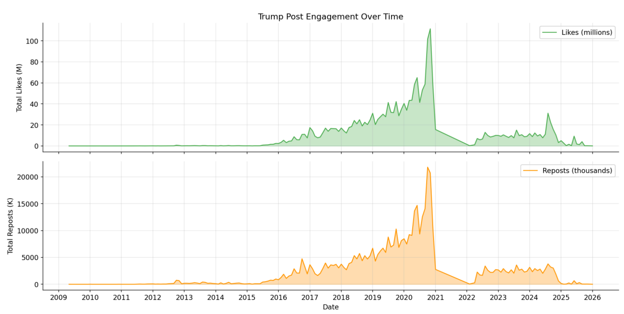

Figure 6: Monthly aggregated engagement (likes and reposts) for Trump's original posts.

Content Analysis

Figure 7 presents a word cloud generated from the lemmatized text of all original posts, with standard English stopwords and common social media tokens removed. Consistent with the preprocessing findings of Kim et al. (2021) for Musk, who found "SpaceX", "NASA", and "TeslaMotors" dominating, Trump's most frequent terms reflect his core rhetorical themes: "great", "president", "country", "america", "election", and "vote" appear prominently, alongside politically charged terms like "democrats" and "fake".

Figure 7: Word cloud generated from lemmatized text of all original Trump posts after stopword removal.

Figure 8 shows the accounts most frequently mentioned in Trump's reply tweets. The dominant handle is @realDonaldTrump, likely self-referential retweet-style replies, followed by @BarackObama and @CelebApprentice, reflecting both political antagonism and Trump's strong association with his TV brand. Media outlets such as @FoxNews and @FoxAndFriends appear prominently alongside critics like @BillMaher and @PiersMorgan, suggesting Trump's replies were largely directed at media figures and political opponents rather than constituents or policy accounts.

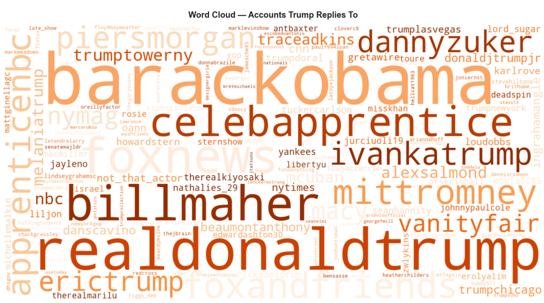

Figure 8: Word cloud of users Trump replied to.

Figure 9 shows topic flag frequencies. The `topic_identity` flag is the most common, appearing in 13,848 posts, followed by elections and rallies nationalism, reflecting the dominance of political identity content in Trump's posting history.

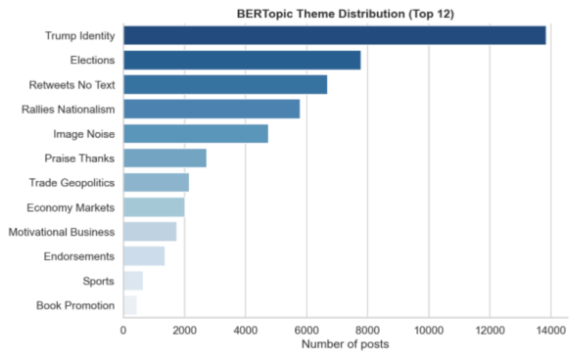

Figure 9: Proportion of original posts flagged for each of the eight economic topic areas.

Semantic Clustering

To examine whether Trump's posts form meaningful semantic groups, we visualised the sentence embedding space using UMAP and coloured each post by its `BERTopic` cluster assignment *(Figure 10)*. The resulting projection shows that several major topics occupy distinguishable regions in the embedding space, including Trump identity, elections, rallies and nationalism, trade geopolitics, and economy-related posts. This suggests that the sentence embeddings capture meaningful thematic differences in the corpus.

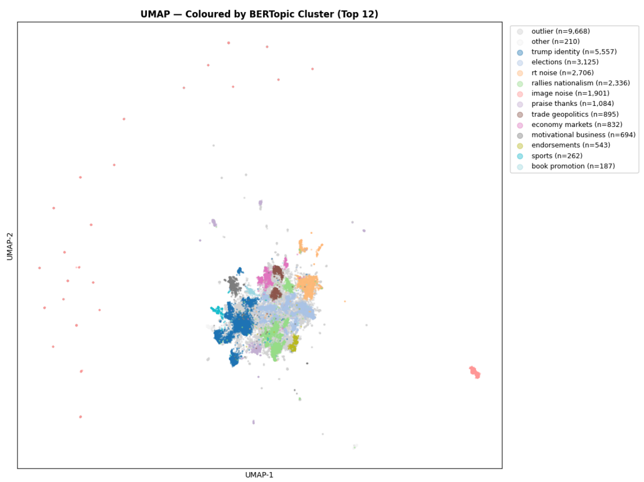

Figure 10: UMAP 2D projection of all 74,742 post embeddings, coloured by BERTopic assigned.

At the same time, the clusters are not perfectly separated. Many posts are concentrated in a dense central region, reflecting the fact that Trump's communication often combines recurring political, personal, and campaign-related language within short social media texts. The presence of outliers and overlapping clusters is therefore expected and supports the use of `BERTopic`, which can assign noisy or weakly coherent posts to an outlier category rather than forcing every post into a clearly defined topic. Overall, the UMAP projection provides a useful visual validation that the embedding-based topic model captures broad semantic structure in the posting corpus, while also showing the thematic overlap inherent in political social media communication.

Two additional UMAP projections are provided in *Figure A2 and Figure A3*. The first colours each post by its next-day S&P 500 return, allowing visual inspection of whether market-moving days correspond to particular regions of the embedding space. The absence of clear colour clustering in this projection is consistent with the weak raw correlations reported in Section 5.7, and reinforces the conclusion that the relationship between semantic content and market returns is not straightforwardly linear. The second projection colours posts by platform — X (Twitter) versus Truth Social — and shows that while the two platforms overlap substantially in embedding space, Truth Social posts are somewhat more concentrated in the identity and rally-related regions, consistent with the platform comparison findings in Section 5.6.

Sentiment Distributions

Figure 11 presents the `VADER` and `FinBERT` sentiment distributions side by side. `VADER` classifies 47.4% of posts as positive, 30.4% as neutral, and 22.2% as negative which is broadly consistent with the Musk distribution found by Kim et al. (2021) of 40% positive and 48% neutral, though Trump's positive share is somewhat higher, likely reflecting his characteristically superlative language. `FinBERT` tells a strikingly different story: 77.6% of posts are classified as neutral, with only

8.8% positive and 13.6% negative. The overall label agreement between the two models is just 46.3%, confirming that the two tools are measuring different dimensions of the text. `VADER` is sensitive to rhetorical intensity, capitalisation, exclamation marks, and words like "great" or "terrible", while `FinBERT`, trained on formal financial news, treats most of Trump's informal social media language as financially neutral. This divergence is important: it suggests that `VADER` captures Trump's rhetorical style while `FinBERT` more conservatively identifies posts with genuine financial content. Taken together, the two models are complementary rather than interchangeable. This also connects to Frydendahl and Stenger's (2019) finding that Trump's company-specific tweets had a positive market effect on the day of posting when sentiment was positive and a delayed negative effect when negative, a result that likely depends on which posts are classified as sentiment-bearing in the first place, and illustrates why the choice of sentiment tool matters.

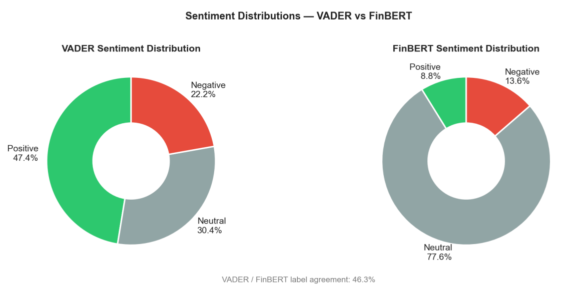

Figure 11: Distribution of VADER sentiment labels across all original Trump posts.

Figure 12 plots the monthly mean `VADER` compound score alongside the cumulative S&P 500 log return from 2009 to 2026. Two features stand out. First, Trump's rolling sentiment is almost always positive. The green shading dominates the chart throughout, with the only sustained negative period occurring briefly around 2011–2012 during his pre-political phase. This persistent positivity is consistent with the floor effect noted in the limitations: `VADER` inflates scores for superlative language regardless of political intent, making the level of sentiment an unreliable signal in isolation. Second, there is no obvious co-movement between the sentiment series and the market trajectory. The S&P 500 rises steadily across the period while sentiment fluctuates without a discernible trend, visually confirming the weak raw correlations reported in Section 5.7. The gap in sentiment around 2021–2022 reflects the Twitter ban and Truth Social transition period. Notably, the market's sharpest drop (the COVID crash of early 2020) occurs without any corresponding collapse in Trump's sentiment score, suggesting that his posting tone was not meaningfully more negative during one of the worst market periods in the dataset. This might be an indicator of mean sentiment as a weak predictor and that dispersion and topic matter more than level.

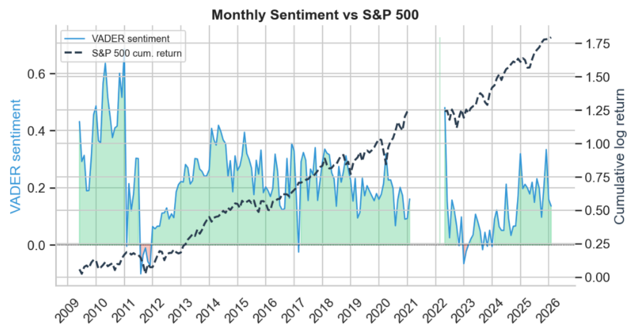

Figure 12: Monthly mean VADER compound score (blue line, left axis) with positive/negative shading, plotted against the S&P 500 cumulative log return (dashed black line, right axis). The gap in sentiment around 2021–2022 reflects Trump's Twitter ban and the transition to Truth Social.

Figure 13 plots the monthly mean FinBERT sentiment score alongside the cumulative S&P 500 log return from 2009 to 2026. The picture differs notably from the VADER equivalent. First, Trump's sentiment is no longer uniformly positive, the FinBERT scores oscillate around zero throughout the entire period, with sustained negative phases in 2011–2012 and again from 2022 onwards. This is consistent with FinBERT's design: trained on financial news, it is less susceptible to the superlative language inflation that inflates VADER scores, and instead captures the adversarial and market-relevant tone of Trump's posts more faithfully. Second, as with VADER, there is no obvious co-movement with the market trajectory. The S&P 500 rises steadily while FinBERT sentiment shows no directional trend, again confirming the weak raw correlations reported in Section 5.7. The gap around 2021–2022 reflects the Twitter ban and Truth Social transition. The spike in early 2021 is notable, FinBERT briefly registers a sharp positive reading immediately before the post-ban silence, potentially capturing a burst of charged financial language around the Capitol period. As with VADER, the COVID crash of early 2020 produces no corresponding collapse in sentiment score, reinforcing the interpretation that mean sentiment level is a weak standalone predictor, and that topic composition and sentiment dispersion are likely more informative signals.

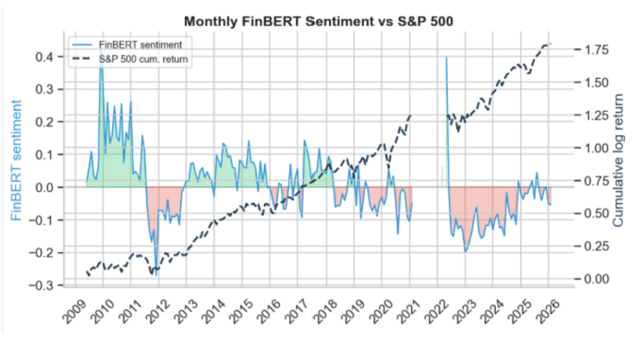

Figure 13: Monthly mean FinBERT compound score.

Platform Comparison

Truth Social posts exhibit meaningfully different characteristics from Twitter posts on several dimensions. *Figure 14* reveals meaningful differences between Trump's posting behaviour on Twitter and Truth Social. On sentiment, both platforms show a similar rightward skew toward positive scores, but Twitter has a notably sharper peak at neutrality while Truth Social shows a somewhat flatter, more dispersed distribution, suggesting slightly more rhetorical variability on his own platform.

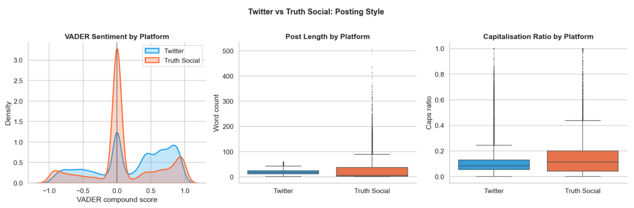

Figure 14: Platform comparison of post characteristics.

Truth Social posts are also longer on average and show a higher capitalisation ratio, indicating a more emphatic writing style.

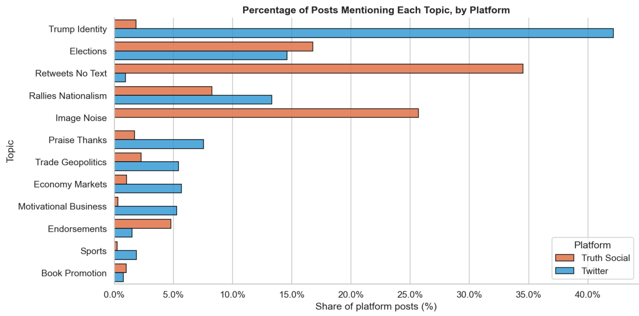

Figure 15: Percentage of posts mentioning each topic, split by platform.

Figure 15 reveals clear differences in topic composition across platforms. Trump Identity dominates Twitter at over 40% of posts versus just 2% on Truth Social, while low-content activity (Retweets, No Text, Image Noise) accounts for roughly 35–25% of Twitter posts but is far less common on Truth Social. Crucially, the financially relevant topics (Trade Geopolitics and Economy Markets) appear at comparable rates on both platforms (2–5%), suggesting that economically relevant posting was not systematically concentrated on either platform.

Sentiment vs Market Returns

Figure 16 presents scatter plots of daily `VADER` mean sentiment against same-day and next-day S&P 500 returns. Both plots show a near-flat OLS trend line and a symmetric cloud of observations with no discernible slope, confirming that raw sentiment level has essentially no linear relationship with market returns at either horizon. This is consistent with Colonescu (2018) and Frydendahl and Stenger (2019), who similarly found limited predictive power from simple sentiment measures. The near-identical appearance of the same-day and next-day plots suggests the absence of signal does not improve with a one-day lag. These findings directly motivate the modelling approach in Section 6: since sentiment level carries little signal on its own, the focus shifts to sentiment dispersion and topic-based features, both of which emerge as the strongest predictors in the permutation importance analysis.

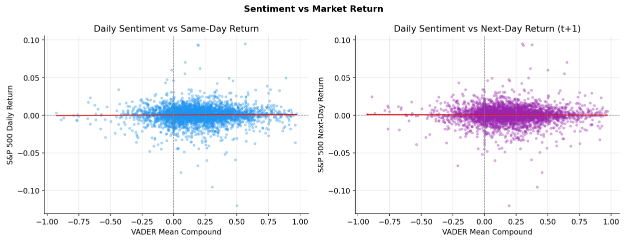

Figure 16: Scatter plots of daily VADER mean compound score vs same-day return (left) and next-day return (right), with OLS trend lines.

The topic impact analysis (*Figure 17*) examines the average next-day S&P 500 return on days when Trump posted about each topic versus days when he did not. The results are heterogeneous across topics. Book promotion, media attacks, and election-related posts are associated with the largest positive differentials, while rallies and nationalism, endorsements, and Trump identity posts are associated with the most negative differentials. The two most financially relevant topics (economy/ markets and trade/ geopolitics) sit close to zero, suggesting that explicitly market- related posts do not systematically predict better or worse next-day returns than average.

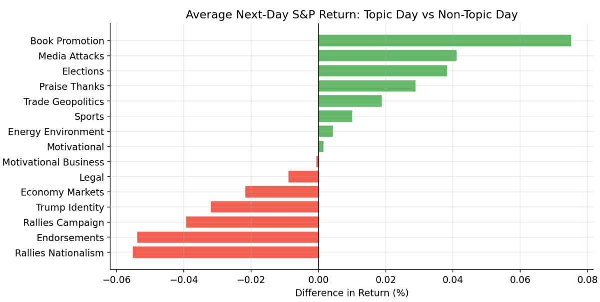

Figure 17: Mean next-day S&P 500 return on days Trump posted about each topic versus days he did not.

Correlation Structure

Figure A4 presents the full Pearson correlation matrix between tweet features and market variables. The most notable pattern is the strong internal correlation among sentiment features such as `vader_mean`, `vader_max_pos`, and `finbert_mean` cluster together, and among market variables such as `log_return`, `next_log_return`, and `abs_return`. Cross-block correlations between tweet features and market variables are uniformly small, consistent with the scatter plots in *Figure 16*.

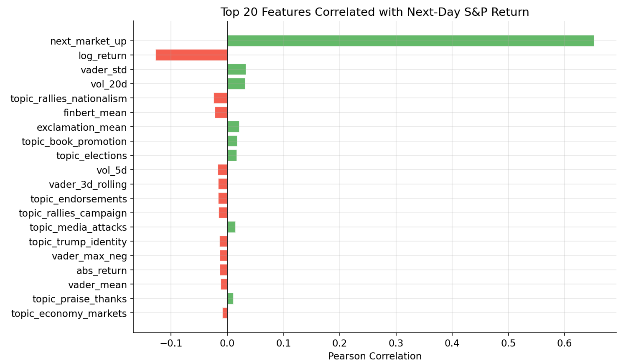

Figure 18: Ranked Pearson correlations of all tweet features with next_log_return.

Figure 18 isolates the 20 features most correlated with `next_log_return`. The dominant signal comes from `next_market_up` and `log_return`, reflecting autocorrelation in market direction, while NLP features such as `vader_std`, `finbert_mean`, and several topic flags show modest but non-zero correlations. Notably, sentiment dispersion (`vader_std`) ranks higher than mean sentiment (`vader_mean`), foreshadowing the permutation importance results in Section 6.

Novelty Score and Event Spotlights

Figure 19 shows the monthly mean novelty score over time. The pattern is highly interpretable. Novelty is highest in 2009–2010, when Trump was new to Twitter and posting about a wide variety of topics such as real estate, sports, business, before settling into a more consistent rhetorical style. It declines steadily through 2013–2019 as his posting became more formulaic and politically focused, reaching its lowest sustained levels during his presidency. A sharp spike appears around 2021–2022, corresponding to the Twitter ban and the disrupted transition to Truth Social, when the embedding baseline was broken by the abrupt change in platform and audience. Novelty rises again from 2023 onward as Truth Social posting becomes more varied. Tellingly, all ten highest-novelty days in the dataset fall between August 2009 and February 2014 with the single highest value recorded on 27 January 2011 (score = 0.897). This confirms that the metric captures genuine semantic discontinuities in posting behaviour rather than arbitrary noise: once Trump's rhetorical style stabilised around his political brand, day-to-day novelty dropped substantially and remained low throughout both presidential terms.

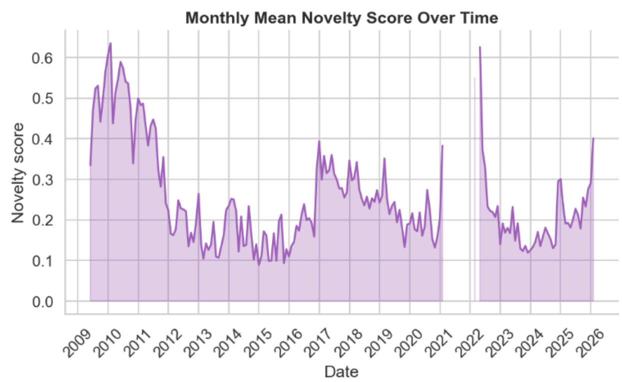

Figure 19: Cosine distance between daily mean post embedding and 14-day rolling baseline. High values indicate semantically unusual posting days.

Although the aggregate analysis reveals no meaningful linear relationship between daily sentiment and returns, zooming into specific periods can reveal episode-level patterns that correlation coefficients obscure. *Figure 20* focuses on the 2018 trade war escalation, overlaying daily `VADER` sentiment with the S&P 500 close price. The period shows sharp alternating sentiment spikes, reflecting Trump's oscillation between threatening and conciliatory trade rhetoric, alongside a sustained market decline from approximately 2,900 to 2,700, consistent with Molnár et al. (2021)'s finding that trade-war tweets had the strongest negative market effect of any topic category. Similar spotlights for the COVID crash and the 2019 Fed rate cut period are presented in *Figure A5 and Figure A6*.

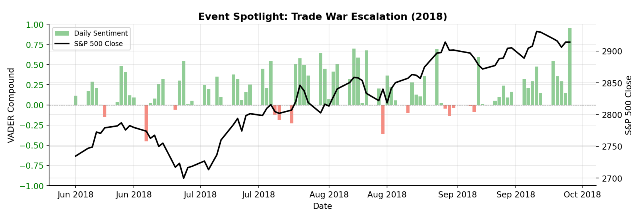

Figure 20: Daily VADER sentiment (bars, left axis) vs S&P 500 close price (line, right axis) during the 2018 trade war escalation.

Finally, *Figure 21* presents an event study of cumulative abnormal returns (CAR) around extreme Trump posting days, defined as days falling in the top and bottom deciles of daily `VADER` compound score. On strongly positive posting days (n=104), the market displays a positive CAR of approximately 0.004 in the days leading up to and including the event, which plateaus around day 0 and gradually declines over the following three trading days.

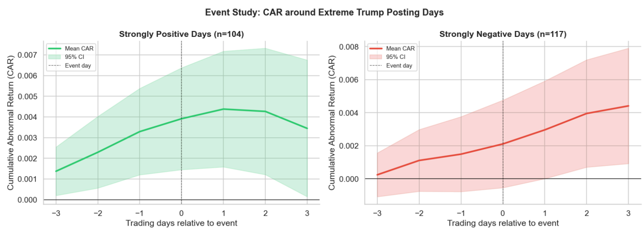

Figure 21: CAR around extreme posting days.

On strongly negative posting days (n=117), the pattern is notably different: CAR is near zero before the event and rises steadily in the days after, reaching approximately 0.004 by day +3. In both cases the 95% confidence intervals are wide and include zero for much of the window, indicating that the effects are not statistically significant at conventional levels. The divergence in post-event trajectories between the two panels is nonetheless suggestive: strongly positive days are associated with a pre-event run-up that fades, while strongly negative days show a delayed positive drift, which is consistent with Frydendahl and Stenger's (2019) finding that negative Trump sentiment has a delayed rather than immediate market effect.

The FinBERT-based event study replicates the same pattern with marginally smaller effect sizes — positive days peak at ~0.003 CAR and negative days show the same delayed upward drift — confirming that the result is robust across sentiment methodologies.

Modelling Strategy

Modelling proceeds in two parallel settings designed to answer distinct research questions. The **next-day setting** asks whether information from Trump's posts on day `t` predicts market outcomes on day `t+1`, using `next_market_up` as the classification target and `next_log_return` as the regression target. The **pre-market setting** asks whether information available before the market opens can predict same-day returns, using only pre-market features, sentiment, post count, and engagement from posts sent between 04:00 and 09:29 ET, alongside lagged market controls. Within each setting we run both classification (market direction) and regression (return magnitude), as the two tasks capture different aspects of predictability.

We use a chronological split rather than a random partition: training covers 2009–2020, validation covers 2022–2023, and the final test period covers 2024–2025. This preserves temporal order and avoids look-ahead bias. `TimeSeriesSplit` cross-validation with five folds is applied within the training period for model selection.

We compare four progressively richer feature sets: (1) a **market-only baseline** using lagged returns and rolling volatility; (2) **market plus simple NLP**, adding `VADER` sentiment, `FinBERT` sentiment, post count, and basic stylistic features; (3) **market plus full NLP**, extending with BERTopic-derived topic flags, caps ratio, and rolling sentiment windows; and (4) **market plus semantic**, additionally including embedding diversity, novelty score, and BERTopic cluster dummies. This incremental design allows us to isolate the marginal contribution of each layer of text features. We evaluate logistic regression, linear SVM, random forest, and gradient boosting classifiers, and ElasticNet and random forest regressors, all fitted through `sklearn` pipelines with imputation and standardisation.

An advanced classification step further tests whether the directional signal becomes clearer when small market movements are removed. We redefine the target to focus on **large next-day market moves**, defined as days where the absolute next-day return exceeds the 75th percentile estimated on the training period. This step uses a tuned walk-forward evaluation scheme, where each test year is predicted using only earlier data. We compare models on all large-move days and on economically relevant event days, defined narrowly as days with at least one post assigned to `topic_trade_geopolitics` or `topic_economy_markets`.

The S&P 500 rises on approximately 53% of trading days in our sample, meaning a majority-class dummy achieves around 53% accuracy. We therefore report balanced accuracy and ROC-AUC as primary classification metrics, since they are less sensitive to class imbalance than raw accuracy.

Modelling Results

Next-day classification. The strongest main classification result is obtained by a Gradient Boosting classifier using the market-plus-semantic feature set. On the final 2024–2025 test set, the model achieves 58.0% accuracy, 54.0% balanced accuracy, and a ROC-AUC of 0.586. This indicates a modest but visible directional signal for next-day S&P 500 movements. However, the validation performance is weaker and closer to random, so the result should be interpreted as limited predictive evidence rather than strong forecasting ability.

Next-day regression. The next-day regression task does not produce meaningful predictive performance. The selected model is the dummy baseline, and the final test R^2 is approximately -0.001. This means that neither market controls nor NLP, topic, or semantic features explain the magnitude of next-day S&P 500 returns better than a naive mean prediction.

The regression null result suggests that Trump's posts may contain weak directional information, but not information about return magnitude. This is plausible given that S&P 500 returns are driven by many factors beyond political social media communication, and is consistent with Frydendahl and Stenger (2019), who find direction effects more detectable than magnitude effects.

Pre-market classification. The best pre-market classification model is a Gradient Boosting classifier using only market controls. On the test set, it reaches 54.4% accuracy, but balanced accuracy is 49.8% and ROC-AUC is 0.497. This indicates that the model does not provide robust same-day directional predictability. The addition of pre-market NLP features does not improve performance, suggesting that early-morning Trump posts do not add reliable information beyond lagged market controls.

Pre-market regression. The pre-market regression task also fails to produce meaningful predictive performance. The selected model is again the dummy baseline, with a test R^2 of approximately -0.001. This confirms that same-day return magnitude is not predictable from pre-market posting features or lagged market controls.

Large-move classification. Large-move classification. The advanced classification setup focuses on large next-day market moves using a tuned walk-forward design. The best all-days model is a Random Forest using the controls-plus-simple-NLP feature set, achieving an overall ROC-AUC of 0.557 and balanced accuracy of 0.564 across 271 predictions and four folds. This suggests that removing small, noisy market movements slightly improves the signal-to-noise ratio. However, the event-days-only model does not provide robust additional evidence. Although its ROC-AUC is similar at 0.558, it is based on only 38 predictions and one fold, with balanced accuracy around 0.497. Therefore, event-day filtering should not be interpreted as a clear improvement.

Feature importance. Permutation importance on the best next-day classification model identifies mean word count as the most important feature, followed by sentiment dispersion measures such as finbert_std and vader_std. Lagged market returns, especially return_lag2 and return_lag1, also contribute to the model. Topic and cluster features have smaller importance values. This suggests that the model's limited predictive signal is driven mainly by text length, sentiment variability, and recent market dynamics rather than by individual topic indicators.

Discussion

Interpretation of Findings

Our results across EDA and modelling tell a consistent but cautious story. The weak raw correlation between sentiment and next-day returns is consistent with Colonescu (2018) and Frydendahl and Stenger (2019), who find that simple sentiment measures have limited predictive power in isolation. However, the modelling results add an important qualification: sentiment dispersion appears more informative than sentiment level. Permutation importance shows that `finbert_std` and `vader_std`, which measure how much sentiment varies across posts within a day, are among the strongest NLP-based predictors. This suggests that the consistency or variability of Trump's posting tone may matter more than the average sentiment score.

The large-move classification results are also informative, but should be interpreted cautiously. Restricting the prediction task to large next-day market movements improves the signal slightly, with the best walk-forward model reaching a ROC-AUC of 0.557. This is broadly consistent with the intuition in Molnár et al. (2021), who find that trade-war-related Trump tweets have particularly strong market effects: text-based signals may be easier to detect when market movements are large and politically salient. However, our results do not provide robust evidence that the signal concentrates specifically on economically relevant event days. Although the event-days-only ROC-AUC is similar, it is based on only 38 predictions and one fold, with balanced accuracy around random performance. Therefore, our findings support the conclusion that large-move filtering improves the signal-to-noise ratio slightly, but not the stronger claim that market-relevant event-day filtering reliably improves predictability.

The novelty score, which measures cosine distance between a day's mean post embedding and its recent posting baseline, is a conceptually original contribution not present in prior studies. Its modest positive contribution is consistent with the intuition that markets may react more to semantic departures from Trump's usual rhetoric than to repeated or predictable language. At the same time, its effect size remains limited.

The structural break introduced by Trump's January 2021 Twitter ban means that models trained across the full period combine different communication regimes. The later test period is dominated by Truth Social and Trump's renewed campaign activity, which raises concerns about the generalisability of patterns learned from earlier Twitter-dominated data. At the same time, any observed performance differences across periods may also reflect broader market conditions rather than platform-specific effects. A comparison with Kim et al. (2021) highlights the limits of the analogy: Musk's tweets operate through a direct single-stock channel, while Trump's macroeconomic and geopolitical posts affect a broad index through more diffuse sentiment and policy uncertainty channels. This diffuseness likely explains why the classification signal remains modest.

Several limitations constrain our findings. First, neither `VADER` nor `FinBERT` is purpose-built for political speech. `VADER`'s lexicon inflates positive scores for Trump's characteristically superlative language ("great", "tremendous", "beautiful" appear even in adversarial posts) while `FinBERT` was trained on formal financial news and may underperform on informal social media text. The fact that sentiment dispersion rather than sentiment level is most predictive may partly reflect these measurement artefacts rather than genuine market dynamics. Second, while `BERTopic` provides a more principled and data-driven approach to topic detection than hand-crafted keyword lists, it is not without limitations. Topic assignments are based on embedding similarity and may group semantically related but economically distinct posts together, for example, trade rhetoric and military threats can co-occur in the same cluster. Additionally, the human labelling step introduces subjectivity, and three topics were left unmapped as noise. Third, the chronological train-validation-test split, while appropriate for avoiding look-ahead bias, means the test period (2024–2025) falls largely within Trump's renewed campaign and is dominated by Truth Social posts. The generalisability of models trained primarily on Twitter data to this regime is uncertain. Fourth, endogeneity remains a fundamental concern. Trump may post more negatively in response to market declines rather than causing them, making the direction of causation impossible to establish from correlational modelling alone. A rigorous causal identification strategy, such as the synthetic control approach used by Li et al. (2023) or a high-frequency event study, would be required to move beyond association. Fifth, the S&P 500 is a broad index affected by countless macroeconomic factors. The signal-to-noise ratio is inherently lower than in single-stock studies such as Frydendahl and Stenger (2019) or Kim et al. (2021), where the posting individual has a direct relationship with the company being tracked.

Conclusion and Future Work

This paper has examined whether features derived from Donald Trump's social media posts, spanning 2009 to 2026 across X and Truth Social, contain information predictive of S&P 500 movements. Our pipeline combined `VADER` and `FinBERT` sentiment scoring, `BERTopic`-derived topic flags, structural stylistic features, 384-dimensional sentence embeddings, and a semantic novelty score, aggregated to a daily feature matrix of approximately 3,526 trading days. Modelling compared market-only baselines against progressively richer NLP feature sets across classification and regression tasks, using a chronological train-validation-test split to preserve temporal integrity.

Our main findings are threefold. First, Trump's posts contain a modest signal for next-day market direction: the best main classification model achieves 58.0% test accuracy, 54.0% balanced accuracy, and a ROC-AUC of 0.586. In the advanced large-move walk-forward setup, the best model reaches a ROC-AUC of 0.557, while event-day filtering does not provide robust additional improvement. Second, sentiment dispersion and text-structure features carry more predictive content than mean sentiment. In particular, the standard deviation of VADER and FinBERT scores within a day ranks above average sentiment in permutation importance. Third, return magnitude is not predictable from Trump's posts under the models examined, with R^2 close to zero or slightly negative across regression experiments.

These findings are broadly consistent with the existing literature. Like Frydendahl and Stenger (2019), we find that direction effects are more detectable than return magnitude effects. Like Molnár et al. (2021), our results are compatible with the idea that politically and economically salient posts may matter more during market-moving episodes, although our event-day-only evidence is too sparse to support a strong conclusion. Like Kim et al. (2021), we find that aggregate posting and sentiment features contain some signal, but that daily market prediction remains noisy and limited.

The two most novel contributions of our study are the platform comparison between X and Truth Social, which reveals systematic differences in posting style and content across Trump's two communication regimes, and the semantic novelty score derived from sentence embeddings, which provides a continuous measure of how semantically unusual a given day's posting is relative to its recent baseline. The novelty measure shows preliminary but limited evidence of predictive value.

Future work should address the causal identification problem more directly, ideally through a high-frequency event study design around specific posts. A sentiment model fine-tuned on political text with financial relevance labels would also likely improve on `VADER` and generic `FinBERT`. Finally, extending the analysis to sector-level indices and individual stocks mentioned by Trump, building on Frydendahl and Stenger (2019), would allow for sharper identification of the channel through which Trump's posts affect financial markets.

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Appendix

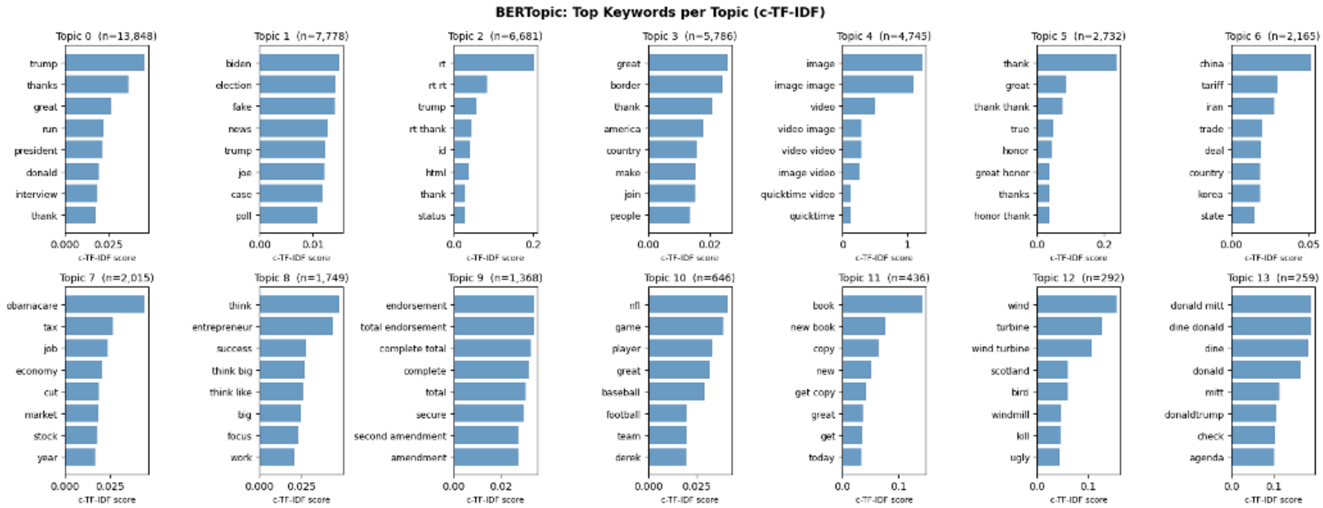

Figure A1 – Most relevant words per topic, assessed by BERT-Topic model.

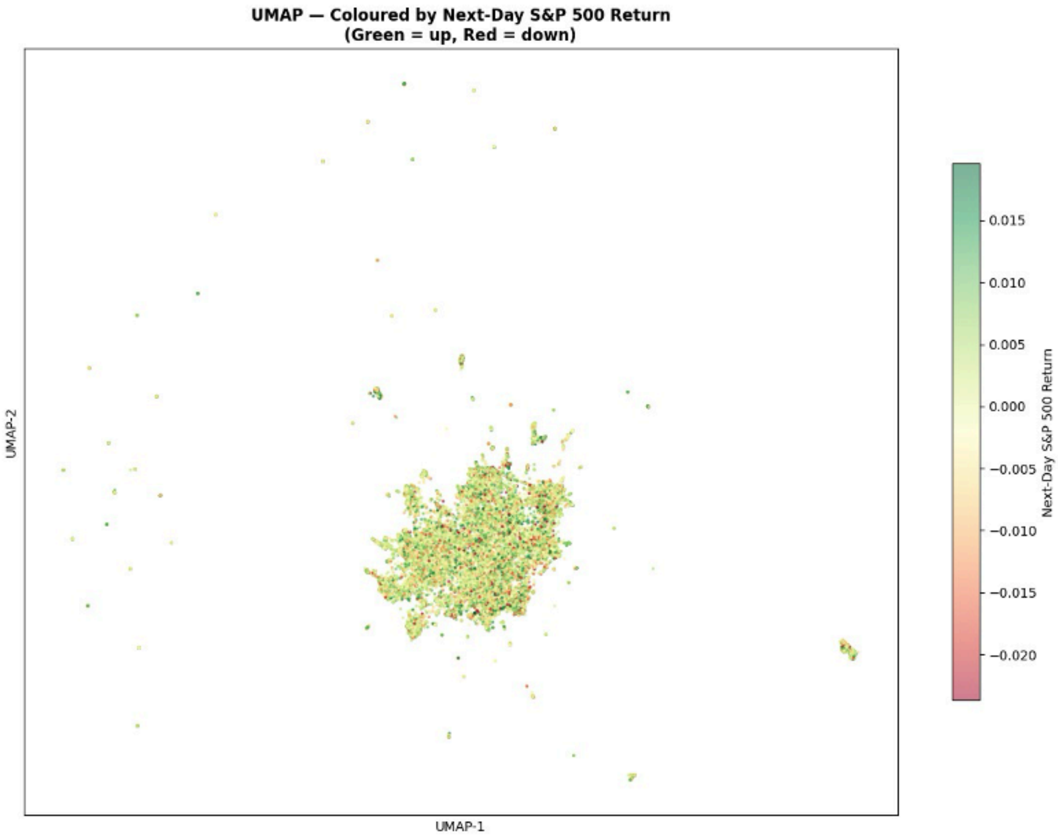

Figure A2 – UMAP projection of S&P500 next-day return.

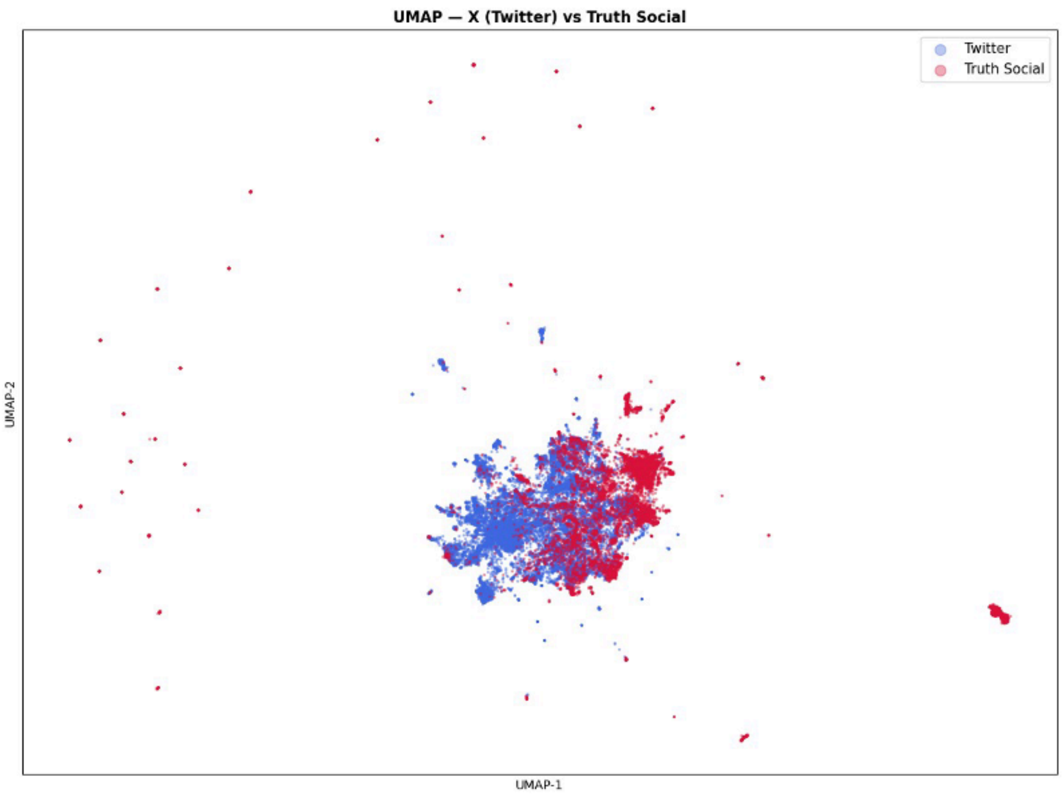

Figure A3 – UMAP Projection of X vs Truth Social.

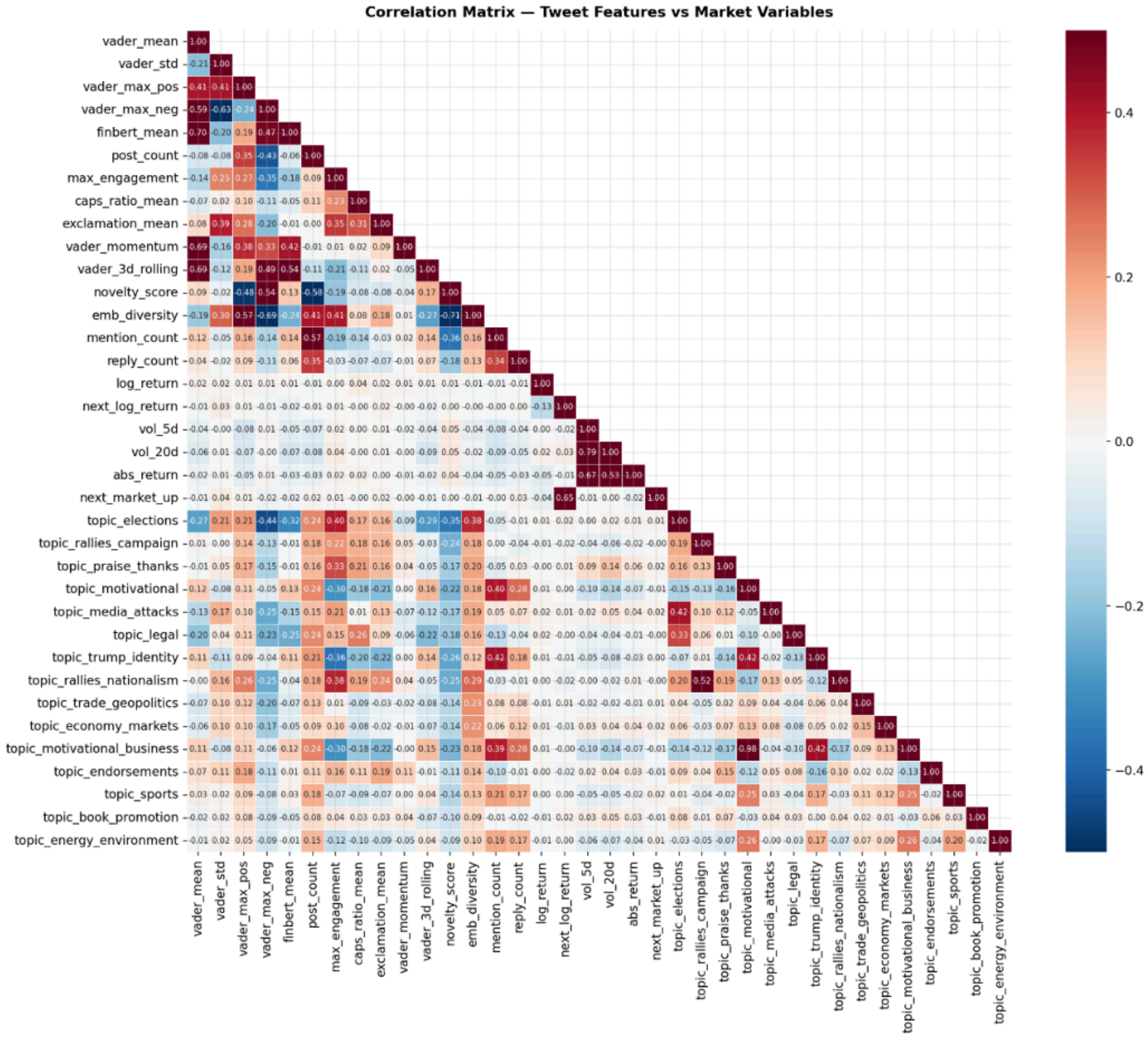

Figure A4 – Pearson correlation matrix of tweet-level features and S&P 500 market variables.

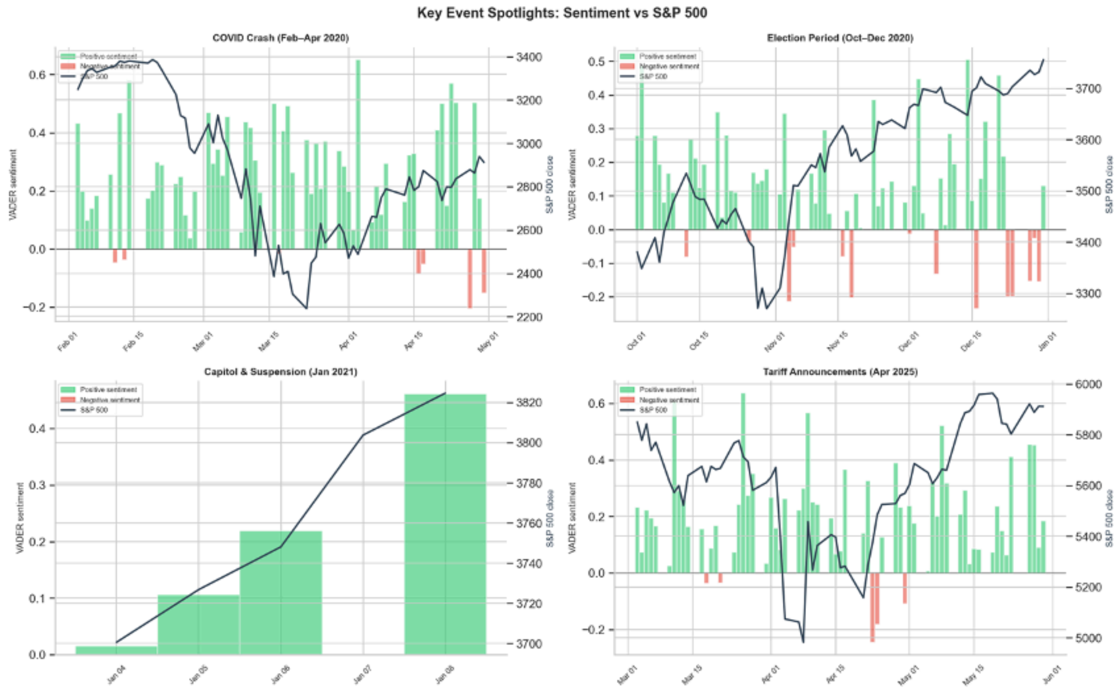

Figure A5 – Vader Sentiment Analysis

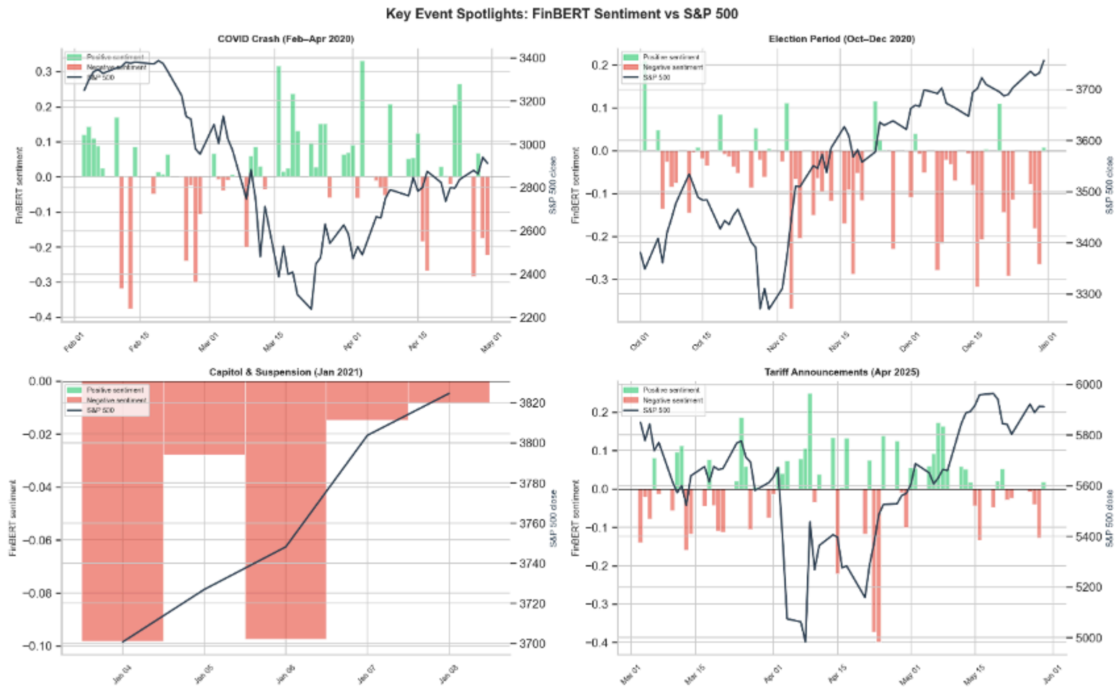

Figure A6 – FinBERT Sentiment Analysis

Key Event Spotlights: Daily VADER (i) and FinBERT (ii) compound score (bars, left axis) and S&P 500 close price (black line, right axis) across four historically significant periods: the COVID crash (February–April 2020), the 2020 election period (October–December 2020), the Capitol riot and Twitter suspension (January 2021), and the April 2025 tariff announcements.

Table A1: Topic keyword flags and representative keyword sets.

Topic Flag	Description	Post Count
topic_trump_identity	Self-referential, personal branding	13,848
topic_elections	Elections, Biden, polls, media narratives	7778
topic_rallies_nationalism	Patriotic rally rhetoric, borders, national pride	5786
topic_praise_thanks	Gratitude, praise, public appreciation	2732
topic_trade_geopolitics	Trade conflicts, tariffs, foreign policy	2165
topic_economy_markets	Economy, taxes, jobs, markets	2015
topic_motivational_business	Business mindset, success, motivational advice	1749
topic_endorsements	Political endorsements, support, Second Amendment	1368
topic_sports	Sports commentary and athlete references	646
topic_book_promotion	Book promotion and sales appeals	436
topic_energy_environment	Criticism of wind energy and environmental themes	292

— Engineered Feature Tables —

Table A2: Preprocessing - Tweet Text

Feature name	Description
date_et	Tweet timestamp converted to US Eastern timezone
post_date	Calendar date of the tweet (date only, no time), in ET
engagement	Sum of favorite_count and repost_count per tweet
tokens	List of cleaned, lowercased, stop-word-free alphabetic tokens extracted from tweet text
token_count	Number of tokens after cleaning
text_preprocessed	Tokens joined into a single string (cleaned version of the text)
pos_tagged	POS-tagged token list (each token paired with its part-of-speech tag)
lemmas	Lemmatized token list, using POS-aware WordNet lemmatization
lemma_count	Number of lemmas after lemmatization
text_lemmatized	Lemmas joined into a single string (used as input for topic modeling and embeddings)

Table A3: Preprocessing – Market Features

Feature name	Description
daily_return	Daily percentage return of the S&P 500 close price
abs_return	Absolute value of the daily return (magnitude of market movement)
next_day_return	Next trading day's return (shifted forward by one row; used as prediction target)

Table A4: Preprocessing - Daily Post Aggregates

Feature name	Description
total_posts	Total number of posts published on a given day
total_engagement	Sum of engagement (likes + reposts) across all posts in a day
avg_engagement	Average engagement per post on a given day
avg_word_count	Average word count across all posts on a given day
twitter_posts	Number of posts published from Twitter on a given day
truth_social_posts	Number of posts published from Truth Social on a given day
quote_posts	Number of quote posts on a given day
repost_posts	Number of reposts on a given day
deleted_posts	Number of deleted posts on a given day
reply_posts	Number of reply posts on a given day

Table A5: Feature Engineering — Tweet-Level Sentiment (VADER)

Feature name	Description
vader_neg	VADER negativity score for the tweet text (0–1)
vader_neu	VADER neutrality score for the tweet text (0–1)
vader_pos	VADER positivity score for the tweet text (0–1)
vader_compound	VADER aggregate sentiment score, ranging from −1 (most negative) to +1 (most positive)
vader_label	Categorical sentiment label derived from vader_compound: positive, neutral, or negative

Table A6: Feature Engineering — Tweet-Level Sentiment (FinBERT)

Feature name	Description
finbert_pos	FinBERT probability of positive financial sentiment
finbert_neg	FinBERT probability of negative financial sentiment
finbert_neu	FinBERT probability of neutral financial sentiment
finbert_label	Dominant FinBERT sentiment class (positive, negative, or neutral)
finbert_polarity	FinBERT polarity score computed as finbert_pos − finbert_neg

Table A7: Feature Engineering — Topic Features (BERTopic)

Feature name	Description
bertopic_id	Numeric topic ID assigned by BERTopic (−1 = outlier/unclustered)
bertopic_label	Human-readable topic label derived from bertopic_id (e.g., topic_elections)
topic_trump_identity	Binary flag: tweet belongs to the "Trump identity" topic
topic_elections	Binary flag: tweet belongs to the "elections" topic
topic_rallies_nationalism	Binary flag: tweet belongs to the "rallies and nationalism" topic
topic_praise_thanks	Binary flag: tweet belongs to the "praise and thanks" topic
topic_trade_geopolitics	Binary flag: tweet belongs to the "trade and geopolitics" topic
topic_economy_markets	Binary flag: tweet belongs to the "economy and markets" topic
topic_motivational_business	Binary flag: tweet belongs to the "motivational and business" topic
topic_endorsements	Binary flag: tweet belongs to the "endorsements" topic
topic_sports	Binary flag: tweet belongs to the "sports" topic
topic_book_promotion	Binary flag: tweet belongs to the "book promotion" topic
topic_energy_environment	Binary flag: tweet belongs to the "energy and environment" topic
topic_mitt_dinner	Binary flag: tweet belongs to the "Dinner with Mitt Romney" topic
rt_noise	Binary flag: retweets (reposts)
image_noise	Binary flag: tweets that act as image legends

Table A8: Feature Engineering — Structural / Stylistic Features

Feature name	Description
hour	Hour of the day the tweet was posted (US Eastern time, 0–23)
weekday	Day of the week (0 = Monday, 6 = Sunday)
caps_ratio	Fraction of alphabetic characters written in uppercase
exclamation_count	Number of exclamation marks (!) in the tweet
question_count	Number of question marks (?) in the tweet
url_flag	Binary flag: tweet contains at least one URL
log_engagement	Log-transformed engagement score: log1p(favorites + reposts)
is_premarket	Binary flag: tweet was posted between 16:00 and 9:29 ET (pre-market hours)
is_market_hours	Binary flag: tweet was posted during regular market hours (9:30–16:00 ET)
is_afterhours	Binary flag: tweet was posted after 16:00 ET (after market close)